

Stat - 486

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Home-Work-4

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Problem 1 & 2 →

4x4 image →

$$\begin{pmatrix} 1 & A & 0 & 1 \\ 2 & 1 & B & 3 \\ 0 & C & -1 & 0 \\ 4 & 0 & -1 & D \end{pmatrix}$$

For any 2x2 patch

$$\begin{matrix} \text{img}[i][j] & \text{img}[i][j+1] \\ \text{img}[i+1][j] & \text{img}[i+1][j+1] \end{matrix}$$

Multiply with ~~filter~~ filter

$$\begin{matrix} \text{img}[i][j] * 1 & + & \text{img}[i][j+1] * (-1) \\ \text{img}[i+1][j] * 0 & + & \text{img}[i+1][j+1] * 1 \end{matrix}$$

$$\Rightarrow \text{out}[i][j] = \text{img}[i][j] - \text{img}[i][j+1] + \text{img}[i+1][j+1]$$

Finding A

$$\begin{matrix} 1 & A \\ 2 & 1 \end{matrix} \Rightarrow 1 - A + 1 = 1 \Rightarrow A = 1$$

Find B

$$\begin{matrix} A & 0 \\ 1 & B \end{matrix} \Rightarrow A - 0 + B = 2 \Rightarrow B = 1$$

Find C

$$\begin{array}{cc} 2 & 1 \\ 0 & C \end{array} \Rightarrow 2 - 1 + C = 3 \Rightarrow C = 2$$

Find D

$$\begin{array}{cc} 1 & 0 \\ -1 & D \end{array} \Rightarrow (1 - 0 + D = 3) \Rightarrow D = 2$$

Values are A = 1, B = 1, C = 2 & D = 2.

Problem 3)

a) Output Dimensions

$$\text{Formula} \Rightarrow \text{output_size} = \frac{(\text{input_size} - \text{filter_size})}{\text{stride}} + 1$$

Input = 32x32

Filter = 5x5

Stride = 1

No padding
3 filters.

$$\text{Output} = \frac{(32 - 5)}{1} + 1$$

$$= 27 + 1$$

$$= 28$$

Problem 4)

We know that making a model more complex reduces bias but increases variance. When we plot test error against model complexity, we get a U-shaped curve, & the best model lives at the bottom of that curve. However, research has shown that this is not the complete picture. If you continue increasing model complexity far beyond that optimal point, the test error first shoots up sharply & then surprisingly comes back down to low levels again. This pattern of two drops in error is what gives double descent its name. The sharp rise happens when the model is exactly large enough to memorize the training data, making it very fragile & sensitive to noise. After that point, the model becomes so large that it has countless ways to explain the training data and the optimization process naturally settles on the most well behaved & generalizable solution among them.

Problem-5)

The attention mechanism works by allocating each word to gather information for all other words in the sequence based on how relevant they are to each other. This relevance is computed purely through mathematical operations on word vectors, with no regard for where each word vector is, encoding its location in the sequence. This allows the model to factor in both

the meaning & the position of each word when computing attention, making it possible to understand how word order affects meaning.

Problem 6) In-context learning refers to a language model's ability to understand & perform a new task based solely on examples provided within the input prompt, without undergoing any form of training or parameter adjustment. The model reads the examples, identifies the underlying pattern and applies it to a new input, all within a single forward pass. Initially, nothing about the model changes during this process. The weights remain fixed & the examples vanish once the conversation ends. This is fundamentally different from traditional machine learning, where a model must be trained on new data to learn new behaviour. In-context learning allows a frozen model to be repurposed for entirely new tasks simply by crafting the right prompt.